

MBAI5310G AI Programming

Assignment 4

# **Decision Tree Model and Business Interpretation**

**TalentWise Advisory**

---

Prepared by: Nusrat Jahan

Student ID: 101050785

Professor: Zahra Atf

Course: MBAI5310G AI Programming

## 1. Understanding the Business Problem

---

### About the Business

TalentWise Advisory is a mid-sized consulting and business services firm with teams across departments like sales, customer support, operations, finance, marketing, IT, and human resources. The company delivers project-based services to corporate clients and depends heavily on skilled employees who carry client knowledge and project experience. Losing trained employees disrupts ongoing projects and forces the company to spend significant time and money on recruiting and onboarding replacements.

### The Problem Being Solved

The core problem is that employee attrition is being noticed too late. By the time someone resigns, the company has already lost the opportunity to step in and address the issues that were bothering that employee. The business is dealing with higher turnover in certain departments and the cost of replacing staff, both financially and in terms of project continuity, is becoming a serious concern. TalentWise wants to identify employees who are at risk of leaving before they actually make that decision.

### How Machine Learning Helps

Machine learning can help HR managers and department heads decide which employees need a supportive conversation or intervention before it is too late. Instead of waiting for a resignation letter, the model can flag employees who show patterns similar to those who have left in the past. Once flagged, HR can review the situation and decide whether a workload adjustment, a career development discussion, a salary review, or extra manager support might help retain that person. The model acts as an early warning system, not a replacement for human judgment.

### Target Variable and Input Features

The target variable is `Left_Company`. A value of 1 means the employee left and a value of 0 means the employee stayed. The input features cover the employee profile, work history, compensation, workload, development activities, satisfaction signals, and work arrangement details. The `Employee_ID` column was removed before training because it is just a unique label with no predictive value.

## 2. Data Preparation

---

### Dataset Overview

The dataset contains 360 employee records with 17 input features and one target variable. There were no missing values and no duplicate rows, which meant the data arrived in clean condition and required minimal preprocessing. The only cleaning step was removing the `Employee_ID` column before training. One important characteristic of this dataset is a significant class imbalance. Only 68 out of 360 employees left, which is about 19 percent of the data. This makes the classification task more challenging because the model has far fewer examples of attrition to learn from.

## Target Variable Distribution

The chart below shows how the records are split between employees who stayed and employees who left. The imbalance is clear, with 292 employees staying and only 68 leaving. This is an important factor to keep in mind when interpreting the evaluation metrics, particularly recall, because a model trained on imbalanced data tends to favor the majority class.

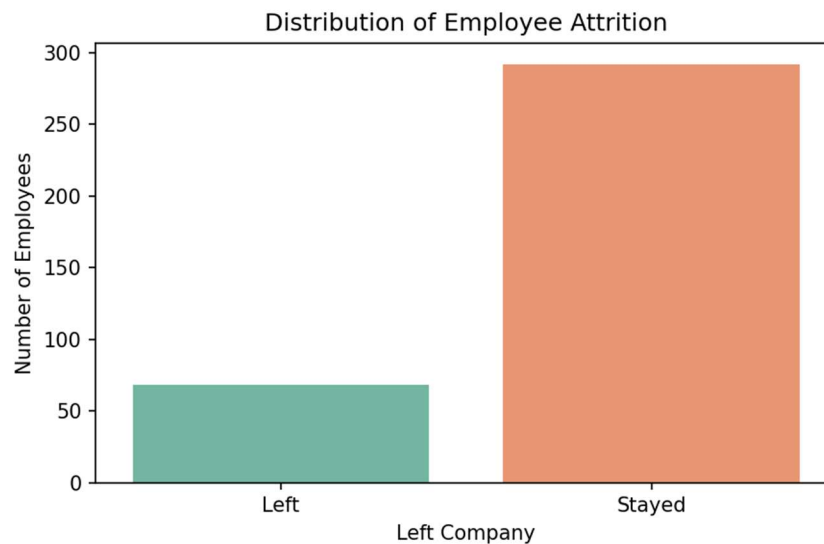


Figure 1: Distribution of the Left\_Company target variable

## Encoding and Splitting

I used one hot encoding to convert categorical columns into numerical format so the Decision Tree could work with them. Each category value became its own binary column. Numerical columns were passed through without any changes. The data was split into 80 percent for training and 20 percent for testing with a fixed random state for reproducibility.

## 3. Training the Decision Tree Model

---

### Model Setup and Entropy

I chose a Decision Tree Classifier for this task because it is well suited for binary classification and produces results that are straightforward to explain in a business context. I set the maximum tree depth to 4 to keep the model simple and reduce the chance of overfitting. I also chose entropy as the splitting criterion instead of the default gini impurity.

Entropy comes from information theory and measures how uncertain or mixed a group is at each decision node. If all employees in a node stayed, entropy is zero because there is no uncertainty. If the group is split evenly, entropy is at its highest. At each step the tree picks the feature and threshold that reduces entropy the most, which is called information gain. Using entropy felt more meaningful for this HR problem because it gives a clearer sense of how well each split is actually separating employees who left from those who stayed.

## Decision Tree Structure

The chart below shows the full structure of the trained tree. Each internal node shows which feature the model is splitting on and what threshold it is using. Each leaf at the bottom shows a final prediction. The color of each node indicates which class is dominant. Reading the tree from top to bottom gives a clear view of the decision logic the model learned.

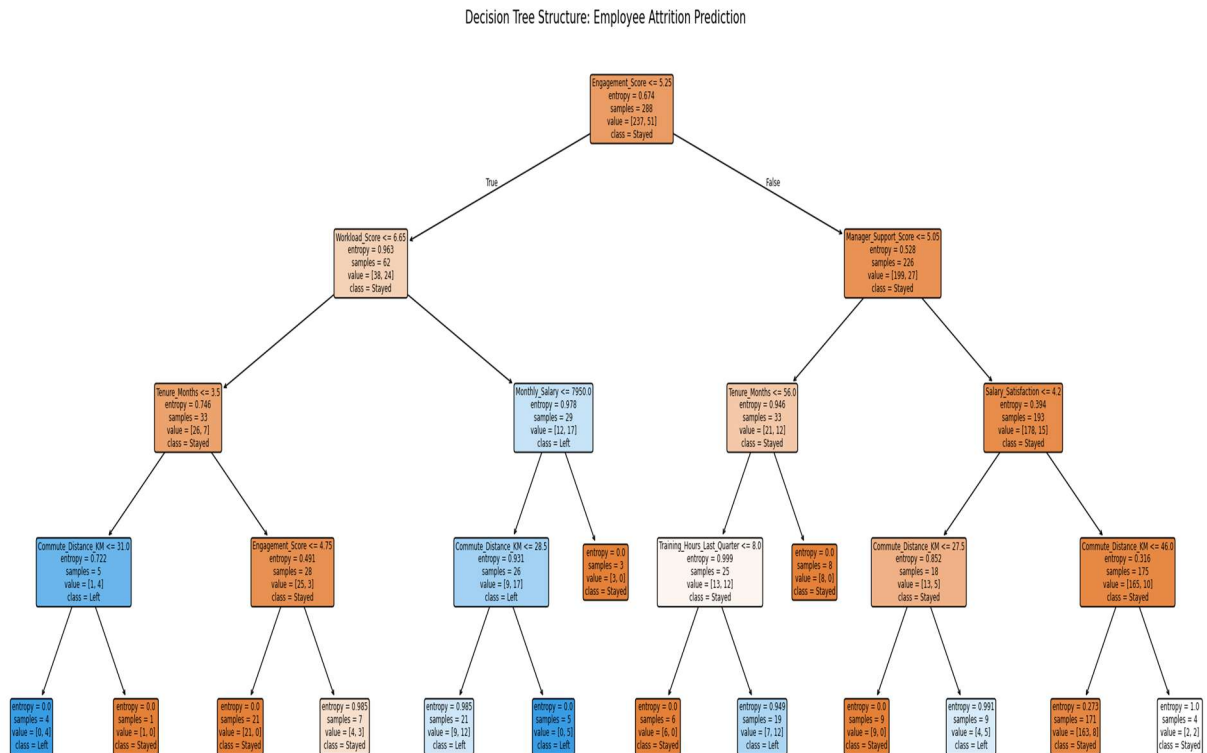


Figure 2: The trained Decision Tree structure with max depth of 4

## 4. Model Evaluation

### Confusion Matrix

The confusion matrix below breaks down the predictions into four categories. True positives are employees who left and were correctly predicted as leaving. True negatives are employees who stayed and were correctly predicted as staying. False positives are employees who stayed but were flagged as likely to leave. False negatives are employees who actually left but were predicted as staying. The false negatives are the most important category to examine for this business problem because they represent missed opportunities to intervene.

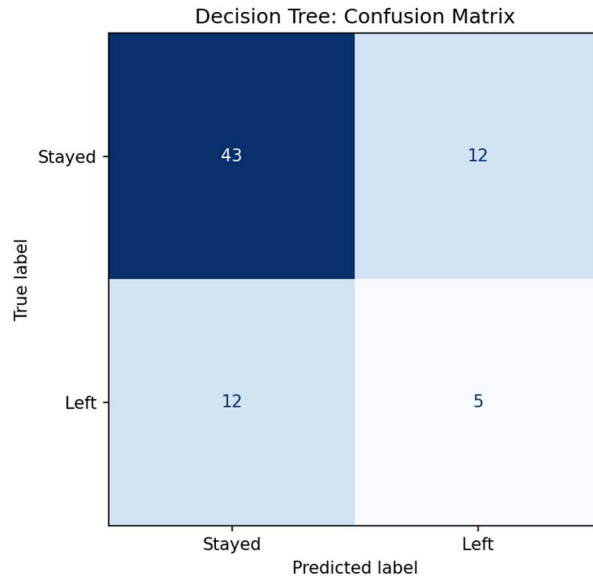


Figure 3: Confusion matrix showing correct and incorrect predictions on the test set

A false positive means the model flagged someone as a flight risk when they were actually fine. The cost is some HR time spent on a retention conversation that was not strictly necessary. A false negative means the model said an employee would stay but they actually left. This is the more costly error because the company had no chance to intervene and bore the full replacement cost.

## Evaluation Metrics

The table below summarizes the key evaluation metrics measured on the test set.

| Metric            | Score  |
|-------------------|--------|
| Training Accuracy | 0.8854 |
| Testing Accuracy  | 0.6667 |
| Precision         | 0.2941 |
| Recall            | 0.2941 |
| F1 Score          | 0.2941 |

## Training vs Testing Accuracy and Overfitting

The chart below compares training and testing accuracy. The training accuracy is 0.8854 and the testing accuracy is 0.6667, giving a gap of approximately 0.22. This is a meaningful gap and indicates that the model is overfitting. It performed well on the data it was trained on but did not generalize effectively to new employee records. Even with a depth limit of 4, the class imbalance made generalization harder because the model had so few attrition examples to learn from. In a production setting, addressing the class imbalance through techniques like class weighting would likely improve generalization.

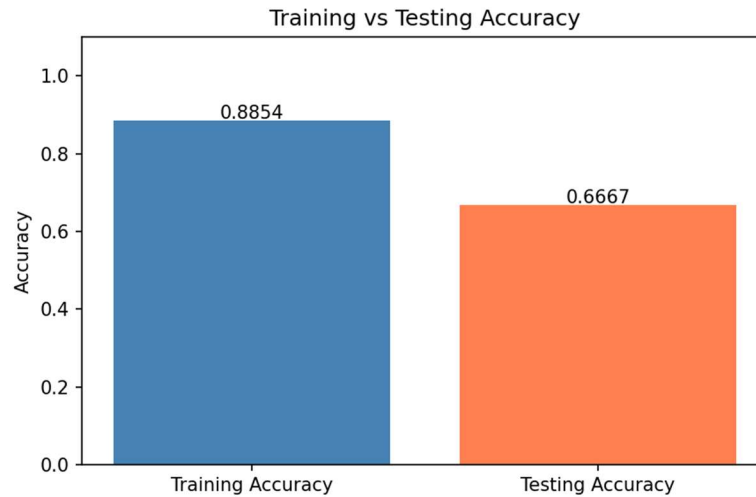


Figure 4: Training accuracy vs testing accuracy comparison

For this business problem, recall is the most important metric. The current recall of 0.29 means the model only caught about 29 percent of the employees who actually left. While this is a weak result, it honestly reflects the challenge of predicting a rare outcome with limited training examples. It also highlights that this model should be seen as a starting point for improvement rather than a finished tool ready for deployment.

## 5. Feature Importance

After measuring model performance, I looked at which features carried the most weight in the predictions. During training, the Decision Tree tracked which features were most useful for separating employees who left from those who stayed. Even though the model showed signs of overfitting, the feature importance scores still provide meaningful directional insight about what drives attrition at TalentWise.

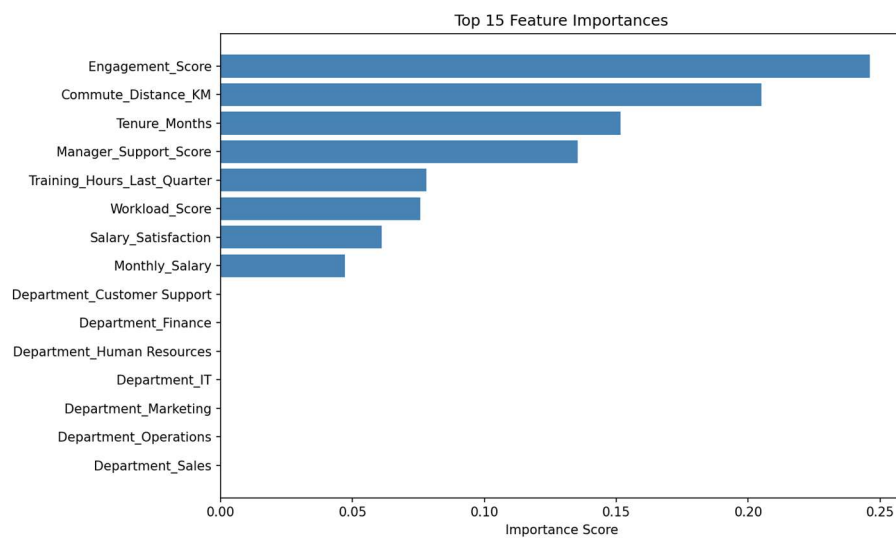


Figure 5: Top 15 most important features from the trained Decision Tree

Engagement\_Score is the top predictor with an importance score of 0.246. This tells me that how connected and motivated an employee feels about their work is the single most valuable signal for predicting whether they will leave. Commute\_Distance\_KM is the second most important feature at 0.205, which suggests that the daily physical burden of a long commute is a meaningful contributor to attrition over time. Tenure\_Months at 0.152 reflects that both newer employees who have not yet settled in and longer tenured employees who feel their career has stalled can be at elevated risk. Manager\_Support\_Score at 0.135 confirms that the relationship between an employee and their direct manager is a key factor in retention.

## 6. Business Insights and Limitations

---

### Key Business Insights

- Engagement\_Score being the top predictor means the company should run regular engagement surveys and monitor score trends at the individual and team level. A declining trend is an early warning that is worth investigating.
- Commute\_Distance\_KM suggests that offering remote work options to employees with long commutes is a retention action with a real business case behind it.
- Manager\_Support\_Score being in the top five means investing in manager coaching and development programs could have a direct positive effect on employee retention.
- Tenure\_Months appearing as an important feature suggests the company should pay attention to both new employees who are still finding their footing and experienced employees who may be feeling underutilized or overlooked for advancement.

### One Limitation of the Model

The most significant limitation is the class imbalance. Only 19 percent of employees in the dataset left, which means the model had far fewer examples of attrition than of employees staying. This directly contributed to the low recall score of 0.29 and the overfitting gap of 0.22 between training and testing accuracy. The model is not reliable enough in its current form to be used as a standalone decision tool. Before any production use, it should be retrained with class balancing techniques, validated on real historical data rather than synthetic records, and regularly monitored for fairness across departments and demographic groups.

### Why Human Judgment Still Matters

Even a well-performing model should never make or imply employment decisions on its own. A flagged employee has their own story and circumstances that no dataset can fully capture. Someone might have high overtime hours because they are excited about a project, not because they are burning out. A low engagement score might reflect a temporary situation that is already improving. Using model predictions without human context risks making interventions that feel intrusive, unfair, or based on incomplete information.

The right way to use this model is as a tool that directs attention. HR managers and department heads should review flagged employees individually, bring their own knowledge of the person and the team context, and decide on supportive actions with care and empathy. The model highlights where to look. The judgment about what to do belongs to the people who understand the employees behind the numbers.